General Relativity BigHW

Problem 1. Prove Bianchi identity

$$\nabla_{[\lambda}R_{\mu\alpha]\nu\beta} = 0$$

Solution:

I will use D as covariant derivative,and work in Riemann normal coordinates at a certain point,that is $\partial_\alpha g_{\mu\nu} = 0 \implies \Gamma^\alpha_{\mu\nu} = 0$.The coordinate can be obtained by diagonalizing metric matrix g,and scaling each eigenvector with the square root of eigenvalues.We denote the coordinates as e_i.The next step is that for all point in the neighborhood,find the geodesic line that connects it and the origin point.Let the tangent vector at the origin be the Riemann normal coordinate of the point.

Directly write out the definition of Riemann tensors:

$$\nabla_{[\lambda}R_{\mu\alpha]\nu\beta} =$$

$$\begin{gathered}
D_\lambda\Gamma^\theta_{\mu\nu}\Gamma^\rho_{\alpha\theta} - D_\lambda\Gamma^\theta_{\alpha\nu}\Gamma^\rho_{\mu\theta} - D_\lambda\partial_\mu(\Gamma^\rho_{\alpha\nu}) + D_\lambda\partial_\alpha(\Gamma^\rho_{\mu\nu}) + \\
D_\alpha\Gamma^\theta_{\lambda\nu}\Gamma^\rho_{\mu\theta} - D_\alpha\Gamma^\theta_{\mu\nu}\Gamma^\rho_{\lambda\theta} - D_\alpha\partial_\lambda(\Gamma^\rho_{\mu\nu}) + D_\alpha\partial_\mu(\Gamma^\rho_{\lambda\nu}) + \\
D_\mu\Gamma^\theta_{\alpha\nu}\Gamma^\rho_{\lambda\theta} - D_\mu\Gamma^\theta_{\lambda\nu}\Gamma^\rho_{\alpha\theta} - D_\mu\partial_\alpha(\Gamma^\rho_{\lambda\nu}) + D_\mu\partial_\lambda(\Gamma^\rho_{\alpha\nu})
\end{gathered}$$

Because we are in Riemann normal coordinates:

$$\nabla_{[\lambda}R_{\mu\alpha]\nu\beta} = -\partial_\lambda[\partial_\mu(\Gamma^\rho_{\alpha\nu})] + \partial_\lambda[\partial_\alpha(\Gamma^\rho_{\mu\nu})] - \partial_\alpha[\partial_\lambda(\Gamma^\rho_{\mu\nu})] + \partial_\alpha[\partial_\mu(\Gamma^\rho_{\lambda\nu})] - \partial_\mu[\partial_\alpha(\Gamma^\rho_{\lambda\nu})] + \partial_\mu[\partial_\lambda(\Gamma^\rho_{\alpha\nu})]$$

And we see directly it is zero because partial derivatives can swap freely.

The source code can be verified in annex BigHw-1.nb.The code was run in Wolfram Mathematica 13.0 with package xact 1.2.0.

Problem 2. Show explicitly that Weyl tensor $W^\mu{}_{\alpha\nu\beta}$ is invariant under conformal transformations such that

$$g_{\mu\nu} \to \hat{g}_{\mu\nu} = \phi(x)g_{\mu\nu}$$

Solution:

$$W^a{}_{bcd} = R^a{}_{bcd} + \frac{R}{(D-1)(D-2)}(g^a{}_c g_{bd} - g^a{}_d g_{bc}) - \frac{1}{D-2}(g^a{}_c R_{bd} - g^a{}_d R_{bc} - g_{bc}g^{ae}R_{ed} + g_{bd}g^{ae}R_{ec})$$

In order the find the weyl tensor after conformal transformation,we firstly calculate some values after conformal transformation.

$$\hat{\Gamma}^a_{bc} = \frac{g^{ad}}{2\phi}(\partial_b(\phi g_{cd}) + \partial_c(\phi g_{bd}) - \partial_d(\phi g_{bc})) = \Gamma^a_{bc} + \frac{1}{2\phi}(g^a{}_c\partial_b\phi + g^a{}_b\partial_c\phi - g^{ad}g_{bc}\partial_d\phi)$$

Let $\phi = e^{\alpha}$

$$\hat{R}^a{}_{bcd} = R^a{}_{bcd} + \partial_{[c} V^a{}_{d]b} + V^a_{[ce} V^e_{d]b}$$

$$= R^a{}_{bcd} + \partial_{[c}(\frac{1}{2}(g^a{}_b \partial_{d]}\alpha + g^a{}_{d]}\partial_b\alpha - g^{ae} g_{d]b}\partial_e\alpha))+$$

$$\frac{1}{4}(g^a{}_e \partial_{[c}\alpha + g^a{}_{[c}\partial_e\alpha - g^{af} g_{[ce}\partial_f\alpha)(g^e{}_b\partial_{d]}\alpha + g^e{}_{d]}\partial_b\alpha - g^{el} g_{d]b}\partial_l\alpha)$$

$$= R^a{}_{bcd} + \frac{1}{2}(g^a{}_d\partial_c\partial_b\alpha - g^a{}_c\partial_b\partial_d\alpha - g^{ae} g_{db}\partial_c\partial_e\alpha + g^{ae} g_{cb}\partial_d\partial_e\alpha)$$

$$+\frac{1}{4}(g^a{}_e g^e{}_d\partial_c\alpha\partial_b\alpha - g^a{}_e g^e{}_c\partial_d\alpha\partial_b\alpha - g^{al} g_{db}\partial_c\alpha\partial_l\alpha + g^{al} g_{cb}\partial_d\alpha\partial_l\alpha)$$

$$+\frac{1}{4}(g^a{}_c g^e{}_b\partial_e\alpha\partial_d\alpha - g^a{}_d g^e{}_b\partial_e\alpha\partial_c\alpha + g^a{}_c g^e{}_d\partial_e\alpha\partial_b\alpha - g^a{}_d g^e{}_c\partial_e\alpha\partial_b\alpha)$$

$$+\frac{1}{4}(-g^a{}_c g^{el} g_{db}\partial_e\alpha\partial_l\alpha + g^a{}_d g^{el} g_{cb}\partial_e\alpha\partial_l\alpha + g_{db}\partial^a\alpha\partial_c\alpha - g_{cb}\partial^a\alpha\partial_d\alpha)$$

$$\hat{R}_{bd} = R_{bd} + \frac{1}{2}(\partial_d\partial_b\alpha - D\partial_d\partial_b\alpha - g_{db}\partial_e\partial^e\alpha + \partial_d\partial_b\alpha)$$

$$+\frac{1}{4}(\partial_b\alpha\partial_d\alpha - D\partial_d\alpha\partial_b\alpha - g_{db}\partial^e\alpha\partial_e\alpha + \partial_b\partial_d\alpha)$$

$$+\frac{1}{4}(D\partial_b\alpha\partial_d\alpha - \partial_b\alpha\partial_d\alpha + D\partial_d\alpha\partial_b\alpha - \partial_d\alpha\partial_b\alpha)$$

$$+\frac{1}{4}(-Dg_{db}\partial^e\alpha\partial_e\alpha + g_{db}\partial_e\alpha\partial^e\alpha + g_{db}\partial_e\alpha\partial^e\alpha - \partial_b\alpha\partial_d\alpha)$$

$$= R_{bd} + \frac{1}{2}((2-D)\partial_b\partial_d\alpha - g_{db}\partial_e\partial^e\alpha) + \frac{1}{4}((D-2)\partial_b\alpha\partial_d\alpha + (2-D)g_{db}\partial_e\alpha\partial^e\alpha)$$

$$\hat{R} = [R + \frac{1}{2}(\partial_e\partial^e\alpha - D\partial_e\partial^e\alpha - D\partial_e\partial^e\alpha + \partial_e\partial^e\alpha)$$

$$+\frac{1}{4}(\partial_e\alpha\partial^e\alpha)(1 - D - D + 1 + D - 1 + D - 1 - D^2 + D + D - 1)]e^{-\alpha}$$

$$= [R + (1-D)\partial_e\partial^e\alpha - \frac{1}{4}\partial_e\alpha\partial^e\alpha(D-1)(D-2)]e^{-\alpha}$$

Thus,the Weyl tensor is

$$\hat{W}^a{}_{bcd} =$$

$$R^a{}_{bcd} + \frac{1}{2}(g^a{}_d\partial_c\partial_b\alpha - g^a{}_c\partial_b\partial_d\alpha - g_{db}\partial_c\partial^a\alpha + g_{cb}\partial_d\partial^a\alpha)$$

$$+\frac{1}{4}(g^a{}_d\partial_c\alpha\partial_b\alpha - g^a{}_c\partial_d\alpha\partial_b\alpha - g_{db}\partial_c\alpha\partial^a\alpha + g_{cb}\partial_d\alpha\partial^a\alpha)$$

$$+\frac{1}{4}(g^a{}_c\partial_b\alpha\partial_d\alpha - g^a{}_d\partial_b\alpha\partial_c\alpha + g^a{}_c\partial_d\alpha\partial_b\alpha - g^a{}_d\partial_c\alpha\partial_b\alpha)$$

$$+\frac{1}{4}(-g^a{}_c g_{db}\partial_e\alpha\partial^e\alpha + g^a{}_d g_{cb}\partial_e\alpha\partial^e\alpha + g_{db}\partial^a\alpha\partial_c\alpha - g_{cb}\partial^a\alpha\partial_d\alpha)$$

$$+\frac{R + (1-D)\partial_e\partial^e\alpha - \frac{1}{4}(D-1)(D-2)\partial_e\alpha\partial^e\alpha}{(D-1)(D-2)}(g^a{}_c g_{bd} - g^a{}_d g_{bc})$$

$$-\frac{1}{D-2}(g^a{}_c R_{bd} - g^a{}_d R_{bc} - g_{bc} g^{ae} R_{ed} + g_{bd} g^{ae} R_{ec}+$$

$$g^a{}_{[c}[\frac{1}{2}((2-D)\partial_b\partial_{d]}\alpha - g_{d]b}\partial_e\partial^e\alpha) + \frac{1}{4}((D-2)\partial_b\alpha\partial_{d]}\alpha + (2-D)g_{d]b}\partial_e\alpha\partial^e\alpha)]$$

$$+g_{b[d} g^{ae}[\frac{1}{2}((2-D)\partial_e\partial_{c]}\alpha - g_{c]e}\partial_f\partial^f\alpha) + \frac{1}{4}((D-2)\partial_e\alpha\partial_{c]}\alpha + (2-D)g_{c]e}\partial_f\alpha\partial^f\alpha)])$$

$$
\begin{aligned}
&= W^a{}_{bcd} + \frac{1}{4}(g^a{}_c g_{bd} - g^a{}_d g_{bc} - g^a{}_c g_{bd} + g^a{}_d g_{bc})\partial_e\alpha\partial^e\alpha \\
&+(g^a{}_d g_{bc} - g^a{}_c g_{bd} + \frac{1}{2}g^a{}_c g_{bd} - \frac{1}{2}g^a{}_d g_{bc} + \frac{1}{2}g^a{}_c g_{bd} - \frac{1}{2}g^a{}_d g_{bc})\frac{\partial_e\partial^e\alpha}{D-2} \\
&+\frac{1}{2}[(g^a{}_d - g^a{}_d)\partial_c\partial_b\alpha + (g^a{}_c - g^a{}_c)\partial_d\partial_b\alpha + (g_{bd} - g_{bd})\partial_c\partial^a\alpha + (g_{bc} - g_{bc})\partial_d\partial^a\alpha] \\
&+\frac{1}{4}[(g^a{}_d - g^a{}_d)\partial_b\alpha\partial_c\alpha + (g^a{}_c - g^a{}_c)\partial_b\alpha\partial_d\alpha + (g_{bd} - g_{bd})\partial^a\alpha\partial_c\alpha + (g_{bc} - g_{bc})\partial^a\alpha\partial_d\alpha] = W^a{}_{bcd}
\end{aligned}
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And we have proved that it is conformal invariant. □